
RESEARCH INTERESTS

LLM post-training (RL, on-policy distillation), LLM agents and agent red-teaming, efficient inference, safety of reasoning models, representation analysis.

EDUCATION

- **University of Wisconsin–Madison** Madison, WI, USA
 - Ph.D. in Statistics; Advisor: Prof. Jun Shao* *Sep 2023 – May 2028 (Expected)*
 - M.S. in Computer Science* *Sep 2023 – May 2026*
 - M.S. in Statistics* *Sep 2022 – May 2023*
 - Visiting Student in Statistics and Computer Science* *Sep 2021 – May 2022*
- **East China Normal University** Shanghai, China
 - B.S. in Statistics* *Sep 2018 – June 2022*

EXPERIENCE

- **SaFo Lab** Johns Hopkins University
 - Research Intern; Mentor: Prof. Chaowei Xiao* *Aug 2025 – Present*
 - Research on safe and efficient large reasoning models and LLM agents (inference-time overthinking mitigation, reasoning distillation, red-teaming).

RESEARCH EXPERIENCE

- **ROM: Real-time Overthinking Mitigation in Large Reasoning Models** UW–Madison
 - First author; advised by Prof. Chaowei Xiao (arXiv 2026)*
 - Framed overthinking in large reasoning models (LRMs) as a latent productive-to-redundant transition, and showed it is a discrete, linearly decodable event in late-layer hidden states around first-correct-solution (FCS) boundaries that survives boundary-permutation and position controls.
 - Built ROM, a streaming framework that monitors a frozen LRM with a lightweight hidden-state detector ($\sim 0.1\%$ of backbone parameters) and intervenes at well-formed reasoning boundaries in real time, with no intermediate-answer extraction, probe decoding, or backbone updates; designed Counterfactual Self-Correction (CSC) to keep useful pre-FCS self-correction while labeling only post-FCS continuation as redundant.
 - Across five backbones from three model families, five reasoning benchmarks, and ten recent baselines under a shared protocol, attained the highest accuracy in 19 of 25 settings, cut response length by 28–77% (mean 45%), and was the only method on the accuracy–length Pareto front in every setting; a single MATH500-trained head transferred zero-shot across scales, families, and domains, cutting wall-clock latency 46.5% at $\sim 5\%$ per-token overhead.
- **ReasoningBomb: A Denial-of-Service Attack on Large Reasoning Models** UW–Madison
 - With Prof. Chaowei Xiao et al. (ACM CCS 2026)*
 - Formalized prompt-induced inference-time denial-of-service (PI-DoS) for LRMs and proved any practical attack must be high-amplification, stealthy, and optimizable.
 - Built a reinforcement-learning attacker, guided by a constant-time surrogate reward ($4.39 \times 10^5 \times$ training speedup), that crafts short, natural prompts driving victim LRMs into pathologically long, often non-terminating reasoning.
 - Across 7 open-source and 3 commercial models, induced $286.7\times$ input-to-output amplification and $6\text{--}7\times$ more tokens than benign queries, with bypass rates of 99.8%, 98.7%, and 98.4% against input-, output-, and dual-stage detectors.
- **PW-OPSD: Position-Weighted On-Policy Self-Distillation for Reasoning** UW–Madison
 - With Prof. Chaowei Xiao et al. (arXiv 2026)*
 - Built a branch-viability diagnostic that forces privileged-teacher next-token alternatives after the student’s on-policy prefix and tests whether the continuation still recovers the correct answer, making teacher-token reliability directly measurable.
 - Showed on Qwen3-4B that an oriented within-sequence position score predicts teacher-token reliability far better than local uncertainty (AUROC 0.83, 95% cluster-bootstrap CI [0.66, 0.95], vs. at most 0.57), i.e. reliability in reasoning distillation is trajectory-structured rather than entropy-driven.

- Proposed PW-OPSD, which up-weights later tokens under the same student rollout, teacher pass, and clipped forward-KL target as OPSD, improving AIME 2024/2025 Avg@12 by +1.0/+1.1 points at no extra teacher compute, with consistent gains on DeepSeek-R1-Distill-Llama-8B and Olmo-3-7B-Think.

• MLE with Datasets from Populations Having Shared Parameters

UW–Madison

Advisor: Prof. Jun Shao (Statistical Theory and Related Fields, 2023)

- Studied maximum likelihood estimation with multiple datasets from heterogeneous populations sharing parameters; established asymptotic normality, estimated variability via bootstrap, and reduced the estimator’s standard deviation by nearly 57% versus single-dataset MLEs.

PUBLICATIONS

Xiaogeng Liu, **Xinyan Wang**, Yechao Zhang, Sanjay Kariyappa, Chong Xiang, Muhao Chen, G. Edward Suh, Chaowei Xiao. *ReasoningBomb: A Stealthy Denial-of-Service Attack by Inducing Pathologically Long Reasoning in Large Reasoning Models*. ACM Conference on Computer and Communications Security (CCS), 2026.

Xiaogeng Liu, **Xinyan Wang**, Yingzi Ma, Yechao Zhang, Chaowei Xiao. *When Are Teacher Tokens Reliable? Position-Weighted On-Policy Self-Distillation for Reasoning*. arXiv:2605.21606, 2026.

Xinyan Wang, Xiaogeng Liu, Ming Pei, Chaowei Xiao. *ROM: Real-time Overthinking Mitigation via Streaming Detection and Intervention*. arXiv:2603.22016, 2026.

Jun Shao, **Xinyan Wang**. *MLE with Datasets from Populations Having Shared Parameters*. Statistical Theory and Related Fields, 7(3):213–222, 2023.

TEACHING

STAT 240 (Data Science Modeling I), STAT 309 (Introduction to Probability and Mathematical Statistics I), STAT 340 (Data Science Modeling II), STAT 610 (Statistical Inference), STAT 611 (Statistical Models for Data Science), STAT 613 (Statistical Methods for Data Science).

PROFESSIONAL SERVICE

Reviewer: ACL 2026, ECCV 2026, EMNLP 2026, NeurIPS 2026

SKILLS

Languages: Python, R, SQL

Frameworks: PyTorch, Scikit-learn, Flask